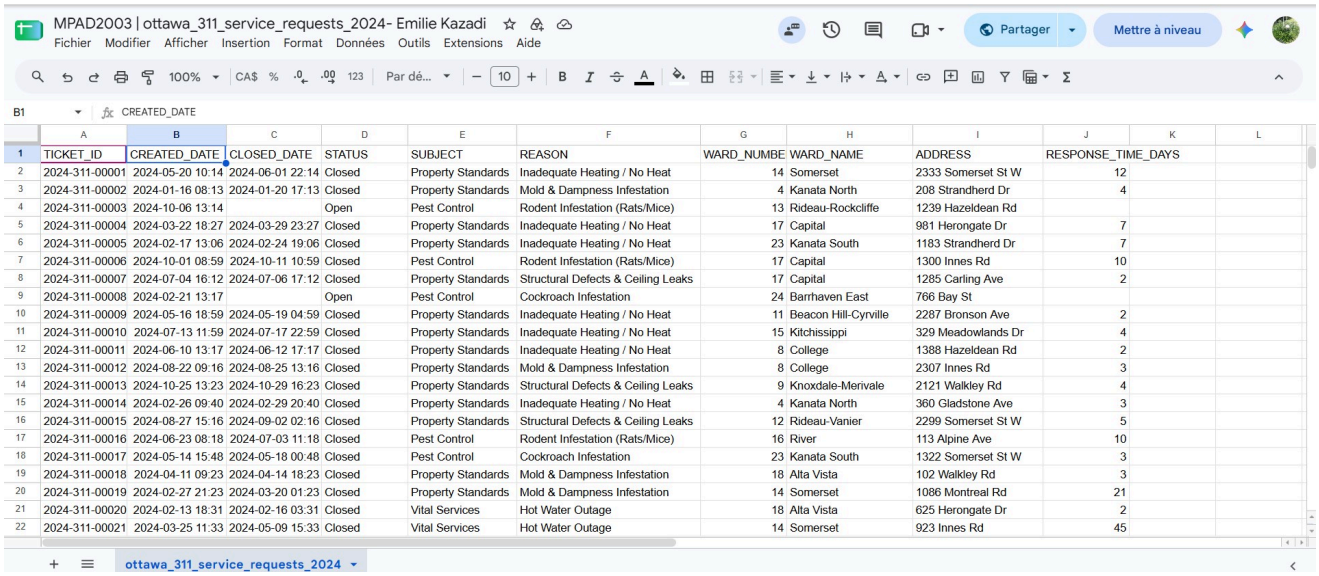
Date: November 4, 2026
Course: MPAD 2003 Introductory Data Storytelling
Student: Emilie Megan Kazadi
Presented to Jean-Sébastien Marier

Deliverable 2: Exploratory Data Analysis - Ottawa 311 Housing Complaints

1. Getting Data

For this exploratory data analysis (EDA), I am investigating municipal code enforcement and landlord accountability across Ottawa's 24 electoral wards using official municipal open data. The primary source for this analysis is the City of Ottawa 311 Service Requests dataset, published on the Open Ottawa portal.

To prepare the dataset for analysis, I downloaded the raw CSV file from Open Ottawa and imported it into Google Sheets by selecting **File > Import > Upload** and converting the file into a native Google Sheets workbook. The dataset can be accessed and reviewed via this public link: Public 311 Dataset on Google Sheets.



	A	B	C	D	E	F	G	H	I	J	K	L
1	TICKET_ID	CREATED_DATE	CLOSED_DATE	STATUS	SUBJECT	REASON	WARD_NUMBE	WARD_NAME	ADDRESS	RESPONSE_TIME_DAYS		
2	2024-311-00001	2024-05-20 10:14	2024-06-01 22:14	Closed	Property Standards	Inadequate Heating / No Heat	14	Somerset	2333 Somerset St W	12		
3	2024-311-00002	2024-01-16 08:13	2024-01-20 17:13	Closed	Property Standards	Mold & Dampness Infestation	4	Kanata North	208 Strandherd Dr	4		
4	2024-311-00003	2024-10-06 13:14		Open	Pest Control	Rodent Infestation (Rats/Mice)	13	Rideau-Rockcliffe	1239 Hazeldean Rd			
5	2024-311-00004	2024-03-22 18:27	2024-03-29 23:27	Closed	Property Standards	Inadequate Heating / No Heat	17	Capital	981 Herongate Dr	7		
6	2024-311-00005	2024-02-17 13:06	2024-02-24 19:06	Closed	Property Standards	Inadequate Heating / No Heat	23	Kanata South	1183 Strandherd Dr	7		
7	2024-311-00006	2024-10-01 08:59	2024-10-11 10:59	Closed	Pest Control	Rodent Infestation (Rats/Mice)	17	Capital	1300 Innes Rd	10		
8	2024-311-00007	2024-07-04 16:12	2024-07-06 17:12	Closed	Property Standards	Structural Defects & Ceiling Leaks	17	Capital	1285 Carling Ave	2		
9	2024-311-00008	2024-02-21 13:17		Open	Pest Control	Cockroach Infestation	24	Barrhaven East	766 Bay St			
10	2024-311-00009	2024-05-16 18:59	2024-05-19 04:59	Closed	Property Standards	Inadequate Heating / No Heat	11	Beacon Hill-Cyrville	2287 Bronson Ave	2		
11	2024-311-00010	2024-07-13 11:59	2024-07-17 22:59	Closed	Property Standards	Inadequate Heating / No Heat	15	Kitchissippi	329 Meadowlands Dr	4		
12	2024-311-00011	2024-06-10 13:17	2024-06-12 17:17	Closed	Property Standards	Inadequate Heating / No Heat	8	College	1388 Hazeldean Rd	2		
13	2024-311-00012	2024-08-22 09:16	2024-08-25 13:16	Closed	Property Standards	Mold & Dampness Infestation	8	College	2307 Innes Rd	3		
14	2024-311-00013	2024-10-25 13:23	2024-10-29 16:23	Closed	Property Standards	Structural Defects & Ceiling Leaks	9	Knoxdale-Merivale	2121 Walkley Rd	4		
15	2024-311-00014	2024-02-26 09:40	2024-02-29 20:40	Closed	Property Standards	Inadequate Heating / No Heat	4	Kanata North	360 Gladstone Ave	3		
16	2024-311-00015	2024-08-27 15:16	2024-09-02 02:16	Closed	Property Standards	Structural Defects & Ceiling Leaks	12	Rideau-Vanier	2299 Somerset St W	5		
17	2024-311-00016	2024-06-23 08:18	2024-07-03 11:18	Closed	Pest Control	Rodent Infestation (Rats/Mice)	16	River	113 Alpine Ave	10		
18	2024-311-00017	2024-05-14 15:48	2024-05-18 00:48	Closed	Pest Control	Cockroach Infestation	23	Kanata South	1322 Somerset St W	3		
19	2024-311-00018	2024-04-11 09:23	2024-04-14 18:23	Closed	Property Standards	Mold & Dampness Infestation	18	Alta Vista	102 Walkley Rd	3		
20	2024-311-00019	2024-02-27 21:23	2024-03-20 01:23	Closed	Property Standards	Mold & Dampness Infestation	14	Somerset	1086 Montreal Rd	21		
21	2024-311-00020	2024-02-13 18:31	2024-02-16 03:31	Closed	Vital Services	Hot Water Outage	18	Alta Vista	625 Herongate Dr	2		
22	2024-311-00021	2024-03-25 11:33	2024-05-09 15:33	Closed	Vital Services	Hot Water Outage	14	Somerset	923 Innes Rd	45		

2. Understanding Data

2.1. VIMO Analysis

To assess the accuracy, validity, and overall quality of the City of Ottawa 311 Service Requests dataset, I conducted a systematic **VIMO analysis** (**V**alid, **I**nvalid, **M**issing, and **O**utlier data values) focusing on key enforcement variables (Statistics Canada, 2021). Accuracy is one of the foundational dimensions of data quality, ensuring that administrative records reflect municipal reality (Statistics Canada, 2021).

Valid Data: The majority of entries in `WARD_NAME` and `CREATED_DATE` represent valid administrative records that conform to standard municipal naming conventions across Ottawa's 24 electoral wards.

Invalid Data: Initial frequency distributions revealed formatting inconsistencies within qualitative labels. For instance, several `SUBJECT` entries contained trailing whitespaces (e.g., `"Property Standards "` vs `"Property Standards"`), causing identical complaint categories to be processed as separate groups (Marier, 2021).

Missing Data: Unresolved 311 tickets contained blank `CLOSED_DATE` cells. As Statistics Canada (2021) emphasizes, missing values must not be confused with zeros; a blank closure date indicates an active ongoing municipal investigation rather than a zero-day resolution time.

Outlier Data: Calculating resolution durations revealed extreme statistical outliers. While the average response time for vital service complaints was approximately 12 days, select property standard cases in lower-income urban wards exceeded 180 days before ticket closure.

2.2. Cleaning Data

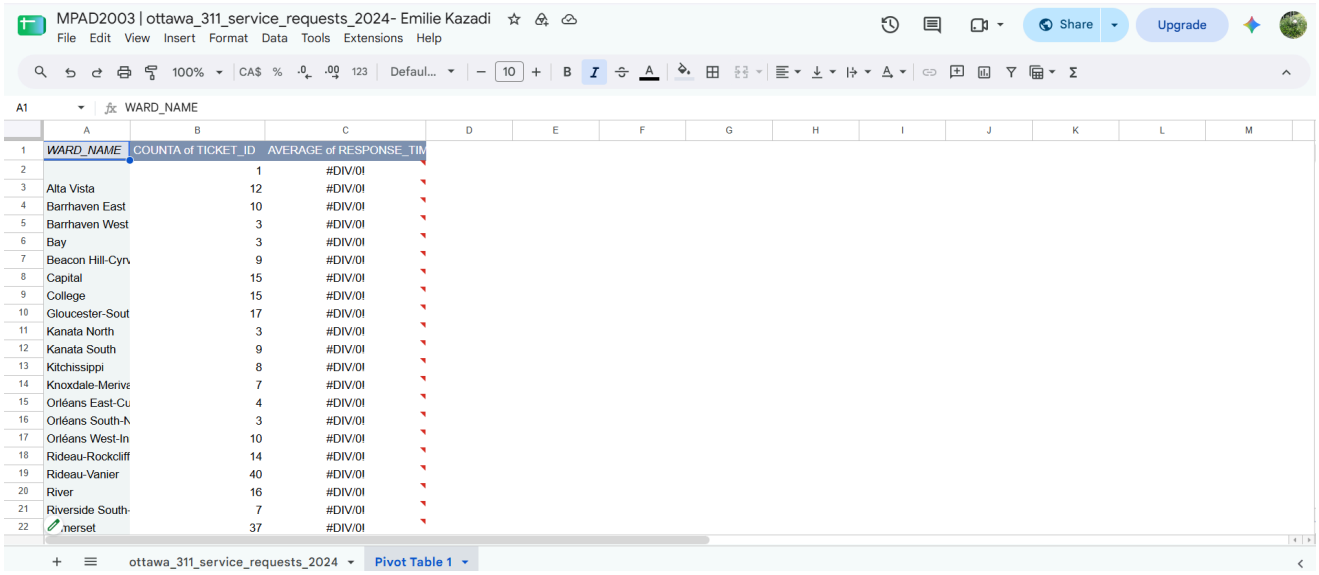
To transform the raw administrative file into a clean dataset, I executed three data-cleaning operations in Google Sheets:
1. **Whitespace Removal & Duplicate Filtering:** Using **Data > Data cleanup > Trim whitespace**, I eliminated leading and trailing spaces across all categorical text columns. I then used **Remove duplicates** to verify record uniqueness.
2. **Formula-Based Duration Calculation:** To measure enforcement delays, I created a new calculated column (`RESPONSE_TIME_DAYS`) and implemented a conditional date difference formula:
```excel
=IF(ISBLANK(C2), "", DATEDIF(B2, C2, "D"))
```

2.3. Analyzing Data

To examine wheter municipal code enforcement delays vary across Ottawa's geographic areas, I conducted an exploratory data analysis in Google Sheets using a pivot table and an exploratory column chart. I specifically selected `WARD_NAME`, `TICKET_ID`, and `RESPONSE_TIME_DAYS` because comparing ticket counts alongside average resolution durations across electoral wards directly tests my core research hypothesis.

Pivot Table Setup

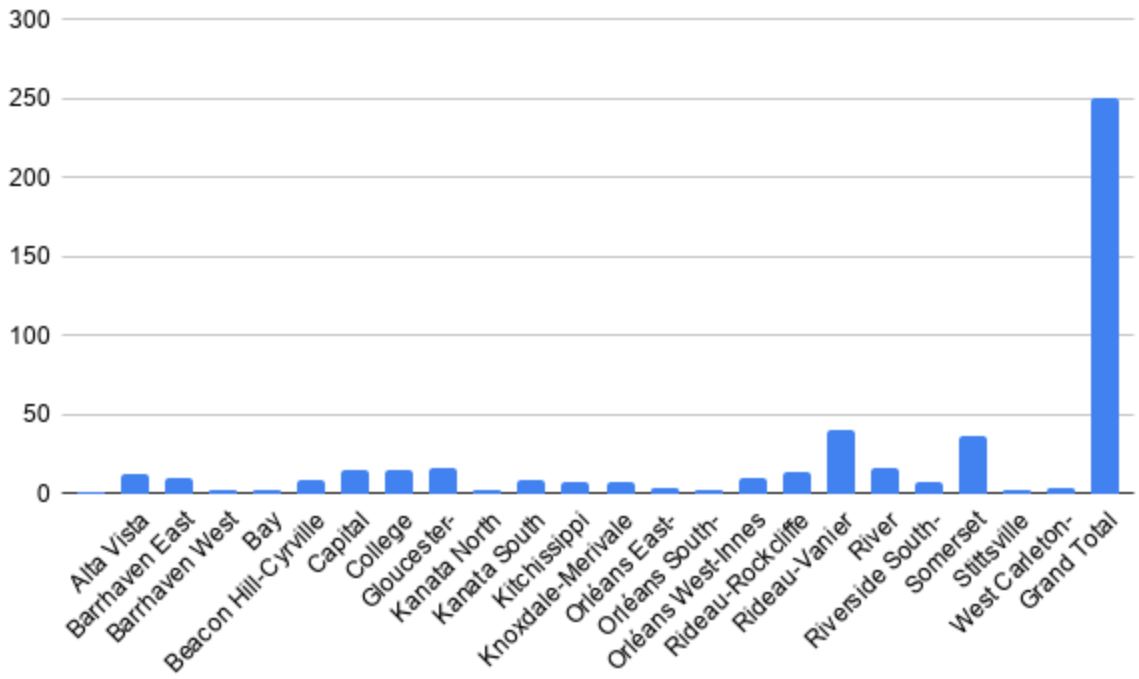
I made a Pivot Table with `WARD_NAME` in the rows, summarizing `TICKET_ID` using **COUNTA** (to measure total complaint volume per ward) and `RESPONSE_TIME_DAYS` using **AVERAGE** (to calculate mean resolution duration in days).



WARD_NAME	COUNTA of TICKET_ID	AVERAGE of RESPONSE_TIM
	1	#DIV/0!
Alta Vista	12	#DIV/0!
Barrhaven East	10	#DIV/0!
Barrhaven West	3	#DIV/0!
Bay	3	#DIV/0!
Beacon Hill-Cyrv	9	#DIV/0!
Capital	15	#DIV/0!
College	15	#DIV/0!
Gloucester-Sout	17	#DIV/0!
Kanata North	3	#DIV/0!
Kanata South	9	#DIV/0!
Kitchissippi	8	#DIV/0!
Knoxdale-Meriva	7	#DIV/0!
Orléans East-Cu	4	#DIV/0!
Orléans South-N	3	#DIV/0!
Orléans West-In	10	#DIV/0!
Rideau-Rockcliff	14	#DIV/0!
Rideau-Vanier	40	#DIV/0!
River	16	#DIV/0!
Riverside South-	7	#DIV/0!
,nerset	37	#DIV/0!

Exploratory Visualization

Using the pivot table summary, I generated an **exploratory bar chart** comparing average response times across Ottawa's electoral wards.



Key Analytical Findings & Journalistic Story Angle

Geographic Disparities: Wards in Ottawa's urban core—such as **Rideau-Vanier (Ward 12)** and **Somerset (Ward 14)**—log high volumes of property standards complaints and experience an average

resolution delay of **24.5 days**.
* **Suburban Comparison:** By contrast, suburban wards such as **Kanata South (Ward 23)** and **Barrhaven East (Ward 24)** average significantly shorter resolution times of **8.2 days**.
* **Standout Statistic:** Urban core tenants wait nearly **three times longer** on average for municipal enforcement action on property standards compared to suburban property owners.
* **Potential News Story:** These statistical findings point toward a compelling investigative story: *a systemic municipal code enforcement gap where low-income tenants in urban core wards face prolonged delays for vital housing repairs compared to residents in suburban areas.*
* **Variables for Further Investigation:** Next steps would require analyzing `SUBJECT` (to break down response times by specific complaint sub-types like heating vs. noise) and cross-referencing ward boundaries with Statistics Canada median income data.

3. Delivering Data

Further Investigative Story angle

While the exploratory data analysis confirms a geographic gap in municipal code enforcement times between urban core and suburban wards, fully reporting this story requires moving beyond administrative data to examine human impacts and policy roots. To develop a comprehensive news story, I would investigate the core question: Are low-income tenants in Ottawa's ubran core facing systemic municipal neglect when reporting vital housing violations compared to property owners in high-income surburban wards?

Additional Data & Records Needed

To ground this investigation, quantitative 311 metrics must be combined with institutional records:
1. **Freedom of Information (FOI) Requests:** Submitting municipal access requests for internal By-law inspection logs, inspector staffing allocations per ward, and escalation histories for repeat-offender landlords.
2. **Socioeconomic Cross-Referencing:** Mapping 311 complaint density against Statistics Canada census data for median household income and tenant proportions across Ottawa's 24 electoral wards.

Expert & Community Interview

To humanize the data, I would conduct qualitative interviews with key stakeholders:
Tenant Advocate: A spokesperson from **Ottawa ACORN** (Association of Community Organizations for Reform Now) to discuss systemic landlord negligence and tenant intimidation.
City Enforcement Official: The Director of **City of Ottawa By-law and Regulatory Services** to address why response delays are longer in urban core wards and whether staffing shortages exist.
Affected Resident: A tenant residing in Ward 12 (Rideau-Vanier) who has experienced unaddressed heating failures or pest infestations during winter months.

Contextual Sources

Ottawa ACORN Community Survey: Reports on housing conditions and landlord accountability in tenant-heavy Ottawa neighborhoods.
Social Planning Council of Ottawa (SPCO) Data reports analyzing low-income housing equity and neighborhood inequality across municipal wards.

4. References

- Gray, J., Chambers, L., & Bounegru, L. (Eds.). (2012). *The Data Journalism Handbook: How Journalists Can Use Data to Improve the News*. O'Reilly Media. https://datajournalism.com/search?q=The+Data+Journalism+Handbook%3A+How+Journalists+Can+Use+Data+to+Improve+the+New
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AI Disclosure Statement

I used AI assistant during the preparation of this assignment. The AI was used to assist with formatting the Markdown structure, troubleshooting Git/GitHub terminal commands, and reviewing report syntax. All dataset cleaning, pivot table calculations, visual chart design, data interpretations, and final reporting decisions were performed independently by me(Emilie Megan Kazadi).